MIL-R-46450 (Ord)
30 December 1960

MILITARY SPECIFICATION

RELAY, M4
LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of relay designated as Relay, M4.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-G-2550 - General Specification for Ammunition Except Small Arms Ammunition.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.

DRAWINGS

ORDNANCE CORPS

7548347 - Box, Set-Up, Packing, Ammunition outer for Relays or Primers. Assembly, Details, Packing and Marking.
7548388 - Box, Packing, Ammunition for Relays, M4, M5 or M7, Assembly, Details, Packing and Marking.
7548389 - Box, Set-Up, Packing, Ammunition, Inner for Relays or Primers. Assembly, Details Packing and Marking.
8844537 - Relay, M4, Loading Assembly.

FSC: 1375

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PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.
GL-7254223 - Gage Lists

(Copies, of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publication should cite "49 CFR 71-78 (latest revision)").

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Relay.-The relay shall comply with all requirements specified on Drawing (dwg.) 8844537, and with all requirements specified in applicable specifications.

3.3 Moisture content of lead azide.-The moisture content of the lead azide used in loading the relay shall not exceed 0.50 percent when tested as specified in 4.3.1.

3.4 Workmanship.-All parts shall be free of chips, dirt, grease, rust, and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. Inspection shall be performed in accordance with this specification and other specifications referenced in any of the contractual documents.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies procured under this specification and shall perform all inspections required by the system. A current written description of the system shall be submitted to the contracting officer, and meet his approval, prior to initiation of production. It shall include a description covering controls for lot formation, and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection (gages, measuring and testing equipment), and provisions for the control and disposition of non-conforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by the provisions of 4.2 and 4.3 and the other documents referenced herein. The contractor shall not be restricted to the inspection station nor the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedure of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that affects the degree of assurance required by this specification or other documents referenced herein.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.
b. Number of units of product inspected.
c. Results obtained for all inspection performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of analyses on all material procured directly by the contractor.
f. Number of items in the lot.
g. Date submitted.

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The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification shall consist of: (a) surveillance of contractor operations to determine that the methods and procedures of the written procedure are being properly applied and (b) Government product inspection. The Government inspector shall also ascertain prior to acceptance that the product complies with all the quality assurance provisions of other specifications referenced in any of the contractual documents.

4.1.3.1 Surveillance.-Surveillance shall include, but is not limited to:

a. Observation of procedures concerning lot formation and identification.

b. Observation of sampling procedures and application of acceptance criteria.

c. Determination that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or approved equivalents thereto.

d. Review of procedures for control and disposition of non-conforming material.

4.1.3.2 Product inspection.-Physical inspection by the Government of contractor accepted product will be performed to the extent necessary to assure contractor compliance with the requirements of this specification and to determine the reliability of contractor records. Severity of Government inspection of individual characteristics will be directly related to the seriousness classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records, Specification MIL-G-2550 will apply.

4.1.3.3 Deviations and deficiencies.-Deviations from the prescribed or agreed upon procedures, or instances of poor practice which might have an effect on the quality of the product, shall be called to the attention of the contractor immediately. To avoid interference, with operation, the contractor shall designate a responsible official, or officials, to whom the Government inspector will report such instances. Failure of the contractor to correct deficiencies promptly shall be cause for suspension of acceptance until correction has been made or until conformance to prescribed criteria has been demonstrated.

4.2 Inspection provisions.

4.2.1. Lot formation.-A lot shall consist of relays produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance, or fit, as

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determined by the Government, shall not necessitate changing the lot number. Each lot shall contain:

a. parts bearing the same interfix lot number from one manufacturer.
b. lead azide from not more than one lot.

4.2.2 Examination.- Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Handbook OHD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Relay assembly (see dwg. 8844537)

Categories	Defects	Method of inspection
Critical:	None defined.	
Major:	AQL 0.65 percent	
101.	Outside diameter, maximum	Gage
102.	Length	Gage
103.	Crimp not full 360°	Visual
104.	Cup damaged	Visual
105.	Charge missing	Visual
106.	Components missing	Visual
Minor:	AQL 0.40 percent	
201.	Evidence of poor workmanship (see 3.4)	Visual

4.2.2.2 Box, Set-up, Inner (see dwg. 7548389)

Categories	Defects	Method of inspection
Critical:	None defined.	
Major:	AQL 0.25 percent	
101.	Contents loose	Manual
Minor:	AQL 1.00 percent	
201.	Label missing	Visual
202.	Marking incorrect, incomplete, or illegible	Visual

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4.2.2.3 Box, Set-up, Packing, Ammunition, outer (see dwg. 7548347)

Categories	Defects	Method of inspection
Critical:	None defined.	
Major:	AQL 0.40 percent	
101.	Contents loose	Manual
102.	Sealing compound incomplete or inadequate	Visual
103.	Waterproof seal broken	Visual
Minor:	AQL 1.00 percent	
201.	Label missing	Visual
202.	Marking, incorrect, incomplete or illegible	Visual

4.2.2.4 Box, packing (see dwg. 7548388)

Categories	Defects	Method of inspection
Critical:	None defined.	
Major:	AQL 0.25 percent	
101.	Strapping missing, broken or loose	Visual-Manual
102.	Box damaged to the extent that contents are exposed or liable to become exposed	Visual
Minor:	AQL 1.50 percent	
201.	Contents loose	Manual
202.	Strapping improperly assembled	Visual
203.	Marking incorrect, incomplete or illegible	Visual

4.2.3 Testing

4.2.3.1 Moisture content of lead azide (see 3.3) Major defect.-The moisture content of each lot shall be determined at the time of loading. A sample of 2 grams shall be selected and tested as specified in 4.3.1. If the moisture content is in excess of the percentage specified, and loading has not begun, the lot of lead azide shall be rejected until proper corrective action has been accomplished as verified by repeating the moisture content test specified in 4.3.1. If loading has begun, action on the unloaded lead azide shall be as specified above and all relays loaded with the questionable lead azide shall be rejected.

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4.2.4 Inspection equipment.-For the performance of the examinations and tests specified in 4.2.2 and 4.3, when applicable, the contractor shall have available, and utilize correctly, all necessary gages, measuring, and testing equipment identified by Gage Lists No. 7254223, or approved equivalents thereto. When inspecting the contractor's inspection equipment, the Government inspector will determine that the contractor has available, and utilizes correctly, gages, measuring, and testing equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. The contractor will have available a set of master gages, standards, and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the contractor and made available for review by the Government. The calibration of gages, standards, and instruments will be periodically checked by authorized Government personnel.

4.3 Test methods and procedures.-

4.3.1 Determination of moisture content of lead azide.-A tared dried aluminum dish 2 inches in diameter shall be covered with a sheet of polytetrofluoroethylene 2 1/2 inches square and sufficiently flat to form a tight cover. An accurately weighed portion of approximately two grams of the sample shall be transferred to a moisture dish and the contents and the dish shall be transferred to an oven and dried at 95 degrees Centigrade (°C.) ± 2° C. for 2 hours. The cover shall be placed in a sulfuric acid desiccator during the drying period. After drying the dish shall be quickly transferred to the desiccator and allowed to cool to room temperature. The polytetrofluoroethylene cover shall be placed on the dish and the assembly weighed. The loss in weight shall be calculated to the percentage of moisture content in the lead azide to determine compliance with 3.3 as follows:

$$\text{Percentage of moisture} = \frac{(A-B)}{A} \times 100$$

where:

A = the weight of the sample before drying
B = the weight of the sample after drying.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level B (see 6.1).-The relay shall be preserved, packaged and marked in accordance with dwgs. 7548347 and 7548389.

5.2 Packing.

5.2.1 Level C.-The relay shall be packed and marked in accordance with dwg. 7548388.

5.3 Data cards.-Data card information shall be as specified in Specification MIL-G-2550.

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6. NOTES

6.1 Ordering data.-Procurement documents should specify the following:
a. Title, number and date of this specification.
b. Selection of applicable level of preservation, packaging and packing
Level B.-In the event the expected storage time of the relay is in excess of 180 days the barrier bag shall be used as specified on the applicable drawing. Expected storage time shall be determined by the procuring agency.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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